

CoE 1195 Senior Project

ChessConnect

Functional Requirements and Engineering Specifications

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Problem Statement

Chess is a game that is traditionally played over-the-board, but as humanity has transitioned into the digital age, the game has undergone many transformations that have affected how players experience it. More and more people are turning to digital chess options with one of the most prominent being the website Lichess.org, which has facilitated over five billion standard rated chess games and just shy of nine billion total chess games since its creation in 2010 [1]. Although the widespread popularity of online chess is undeniable, using these services results in compromises that change the core of the game. Many players find it easy to get distracted while playing chess online due to plagues of notifications, pop-ups, and advertisements which can easily break their focus and lead to less precise play. Furthermore, this shorter attention span is highlighted by the fact that online chess players have a preference to games with shorter time controls. This contrasts with traditional tournaments played over-the-board which often give hours for each side. Many experienced chess players confirm playing chess in the physical world requires a different skill set. In her blog post detailing some of these differences, Estelle Oberholzer, owner of a chess supply shop and creator of the blog “House of Chess”, highlights that most online chess boards are two-dimensional. This means that the recognition skills built online often do not translate well to a physical, three-dimensional board [2]. This coordination is needed for players to effectively maneuver their pieces under pressure given that in chess if a piece is touched it must be used for the next move. This contrasts with online chess wherein players can draw arrows on the board to visualize sequences of moves before they are made. Given these differences, players who are used to online chess can find themselves dealing with a skill-gap when it comes to traditional gameplay, especially in a tournament setting.

Project Description

ChessConnect allows players the easy accessibility of online play, while still allowing them to develop the skills necessary for over-the-board play. ChessConnect links seamlessly with one of the largest internet chess servers, Lichess.org, and allows users to play with any type of opponent on the site including: random players, specific friends, or chess computers. This provides users with the flexibility of online chess while building good habits for over-the-board play. Using an existing online chess service also reduces the higher cost associated with

integrated chess computers and proprietary solutions. ChessConnect further enhances the traditional chess experience by automatically translating the online opponent's moves to the physical board using an internal electromagnet. By interfacing with magnets internal to the pieces, ChessConnect removes the burden of manually updating the pieces seemingly by magic and maintains a traditional look and feel. After the initial setup, ChessConnect only needs to be connected to power and does not require the use of an external application on a personal computer to function. Once the device is connected to Wi-Fi and linked to the user's Lichess.org account, ChessConnect operates using only the physical buttons to start a game. The LCD screen prompts the user to select a game-mode and time controls, as well as shows the chess clock once a match starts. During a game, the screen also shows the destination of the opponent's piece while it is being moved and gives warnings and corrections for illegal moves accompanied by an audible alert. An overall design sketch of ChessConnect is shown in Figure 1.

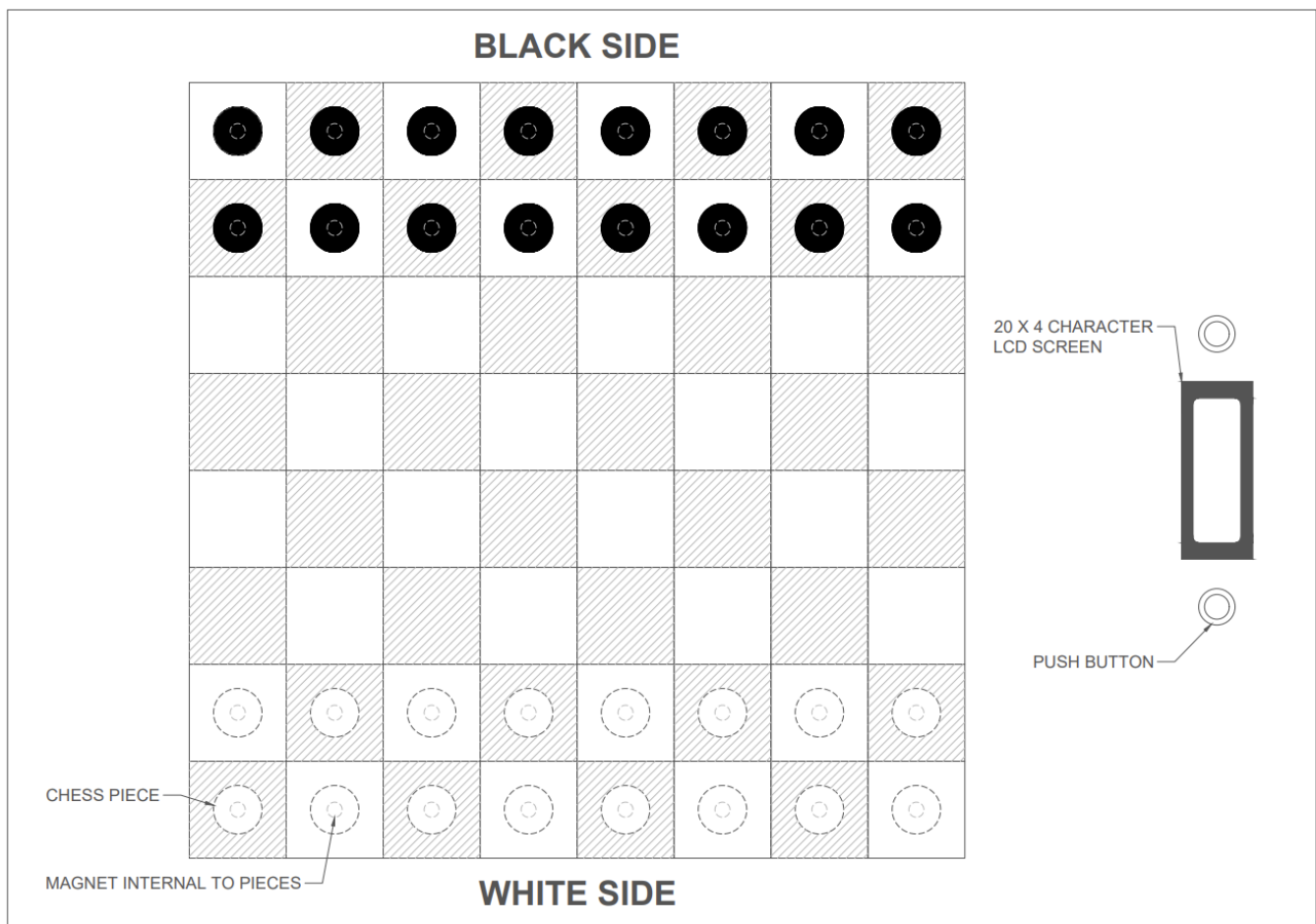


Fig. 1. Overall sketch of ChessConnect.

Throughout a chess game, ChessConnect manages the movement of pieces, verifies their positioning, and communicates with the online game server. In doing so, the board allows for seamless connectivity and gameplay with online opponents. A flowchart describing how ChessConnect functions during a chess game is shown below in Figure 2.

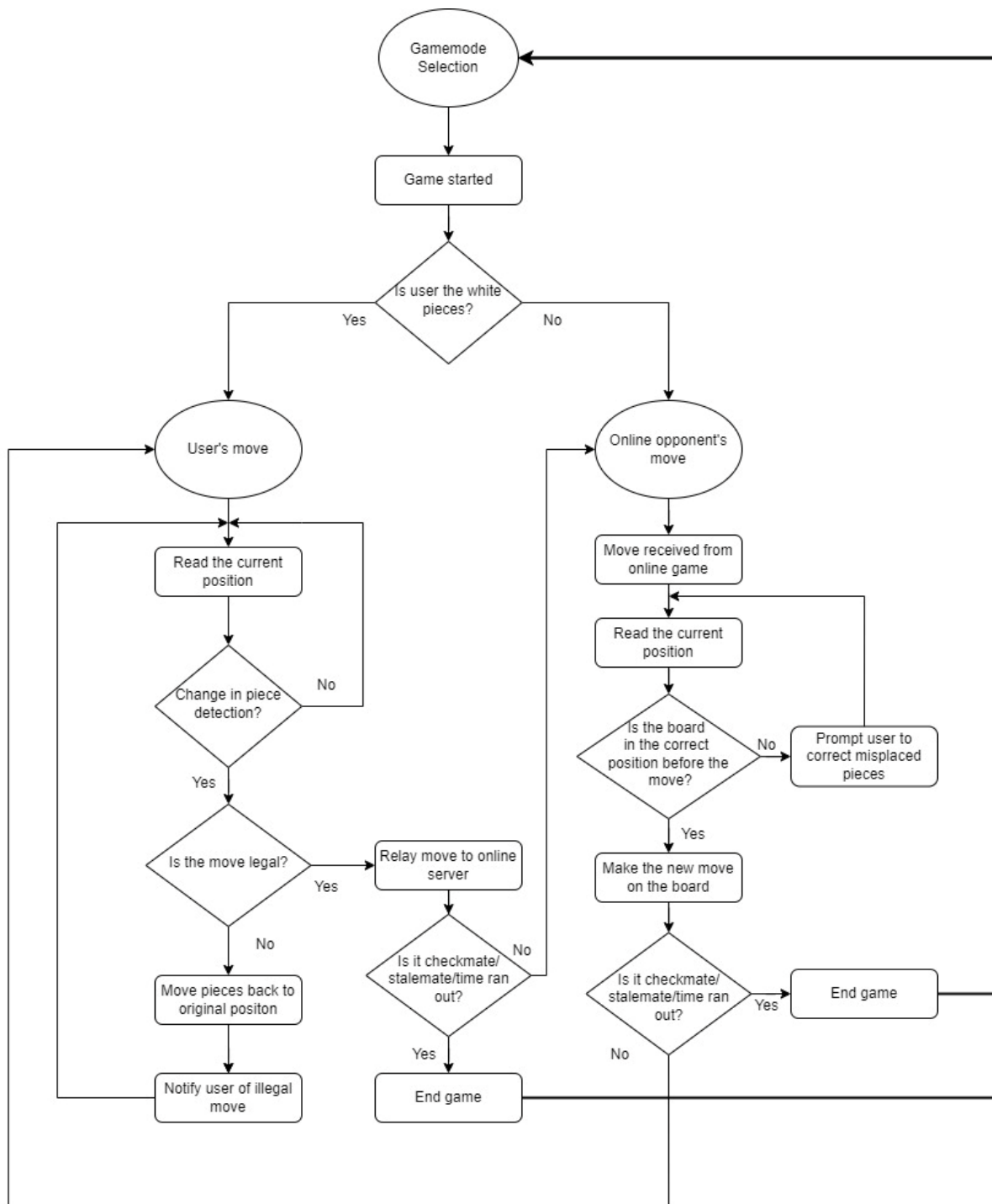


Fig. 2. Flowchart of program control during a chess game.

Functional Requirements

The functional requirements listed below are important qualities that the ChessConnect must possess in order to meet the user's expectations for the product and achieve the desired behavior. These requirements were then incorporated into the House of Quality (HoQ) to highlight the areas that consumers would value most and to visualize their relative importance to one another as shown in Figure 3. The list below highlights each of the requirements and gives a brief explanation of each:

- **Affordability:** how much ChessConnect will cost. If the product is outside of the average customer's price range, it fails at being accessible to a wide audience. Ideally, ChessConnect will have ample features compared to competitors while maintaining a viable price point.
- **Playable Area:** the amount of space allotted for the chessboard on ChessConnect. The board should remain large enough to easily access pieces and visualize the game while also being small enough to allow for portability and fit the standard for chessboard dimensions.
- **Size of Pieces:** the size of the chess pieces to be used on the board. The sizing of pieces is an important requirement for those who wish to practice for tournament play. Pieces must follow guidelines so that users practicing with ChessConnect can easily transition their muscle memory to a tournament. Furthermore, larger chess pieces are typically viewed as being of higher quality and offer a better experience. To offer the best end-user experience, the size of each piece should be as close to chess tournament standards as possible.
- **Build Quality:** the overall quality of the ChessConnect device and experience. The materials should be sturdy enough to withstand the movement of the internal XY rail system for prolonged use and the aesthetics should be of a relatively high quality while maintaining affordability. The gameplay experience should be as seamless as possible, minimizing any errors or connectivity issues.
- **Piece Movement Speed:** the speed at which automatically moved pieces will travel. If a piece is moved too quickly, it may lead to less precise movement and knock over other

pieces. However, if a piece is moved too slowly, it can become an inconvenience for the user and may impact their time remaining to make their next move.

- **Ease of Use:** how comprehensive the user interaction with the device must be. The startup sequence of ChessConnect must be intuitive, allowing users to quickly begin playing. It is necessary to develop the UI so that users can effectively navigate all features of the device while minimizing the amount of selection screens and input required to do so. Along with this, the display on the board must be informative, responsive, and straightforward at all times.
- **Gameplay Options:** how many gameplay modes are available. These gameplay options include different time controls, varying difficulties for online bots and opponents, and even the capability to play a normal game over-the-board. For a customer, having as many gameplay options as possible will make for the best user experience. Minimizing these choices can lead to less customer satisfaction and less functionality.

Engineering Specifications

After functional requirements were determined, engineering specifications were created as a means to measure each functional requirement. These specifications along with their relationships to the functional requirements are detailed in the middle of the HoQ as shown in Figure 3. The positive (+) and negative (-) correlations between each of the engineering specifications are detailed in the roof of the HoQ. The engineering specifications have two values: a design target and a design threshold. The design target is the desired value for a given engineering specification and is deemed the best-case scenario in achieving the desired functionality. The design threshold is the value determined to be the minimum acceptable target for a given specification to fulfill the desired functionality. The design targets and thresholds are shown at the bottom of the HoQ as well as in Table 1. The engineering specifications, along with a brief explanation of their associated design targets and thresholds, are outlined below:

- **Manufactured Cost:** the total cost in USD of the raw components needed to create ChessConnect. The cost target was determined based on the components that have been selected to create the device. The current total of these components is roughly \$300 before taxes, so the target for manufactured costs has been set at \$350. ChessConnect aims to be a more budget-friendly option compared to its competitors, therefore

determining a reasonable threshold for the manufactured cost is critical. Most of the electronic chess boards on the market are priced similarly to the target of \$350, but they also lack the automatic movement functionality that ChessConnect provides.

Additionally, the only electronic chess board currently on the consumer market that matches ChessConnect's functionality, Square Off's Miko Chess Grand, retails for \$550 [3]. Given that this product is ChessConnect's main competitor in terms of features, the cost threshold was set at \$500 to maintain a cheaper cost than the competition. This threshold adds 43% to the manufactured cost target which will account for any costs that may arise due to the addition of components if needed.

- **Size of Device:** how large ChessConnect will be dimensionally in inches (Length x Width x Height). The device must be small enough to be portable and convenient for the end user while being large enough to support a sizeable playable area and allow for the internal electronics. In order to have space for the rail system, LCD screen, push buttons, and other components, the design target is 27in x 19in x 3.5in. This will allow for an extra 11 inches of length and 3 inches of width outside the playable area (16in x 16in). This extra area will house internal components, allow for XY rail mobility, and be used for the LCD display. Along with this, the height of 3.5 inches will provide enough vertical clearance for the rail system and reed switches to be housed inside the board. The design threshold at 32 in x 24in x 4.5in provides increased tolerance for mounting components and the XY rail system.
- **Automatic Piece Move Time:** the time, in seconds, it takes from when ChessConnect receives the opponent's move until the pieces are repositioned on the board. Since chess has a large variance in move types, it can be challenging to find a standard for measuring the time it takes for the device to reflect the virtual opponent's move. For example, moving a pawn one square takes much less time than taking a piece on the other side of the board with a rook. To create a standard for measuring this process, a specific move has been determined as a basis for measuring the time it takes to receive the opponent's move, update the piece positions, and turn over control to the ChessConnect user: a pawn taking a pawn in the center of the board. This move involves moving the taken piece outside of the playable area and moving the opponent's piece to the new space which is a good middle ground between overall move time and move complexity. This move is also

very common in many early game positions given that one of the core principles of opening play is to control the center of the board using pawns. According to Chess.com's explorer, which displays the most common opening moves among master games, the predominant opening moves involve moving several pawns to the center of the board [4]. This results in the very common scenario where one of these pawns captures another. Target and threshold values were determined based on analysis of the only other electronic chessboard that moves pieces automatically, the Miko Chess Grand [3]. Online videos of this board were used to make an estimation for the threshold and target time. The design target for the standard move time is 6 seconds, and the threshold is 11 seconds.

- **Time to Start a Game:** the number of screens required to start a new game. The design target for this specification is 3 screens, and the threshold is 6. These values were determined by estimating the number of necessary screens needed to get into a game with an opponent. The board must have at least a start screen, a screen to select opponent type, and a screen to select time control. While it is best to keep this specification to a minimum, some features may require more options and selection screens. Therefore, the design threshold is set to 6 screens to accommodate these potential occurrences.
- **Piece Detection Errors (per 40 moves):** the number of incorrectly detected pieces for every 40 moves from the user or opponent. A piece detection error occurs whenever the board is unable to accurately detect the piece positions when reading the user's move or verifying the opponent's move. The primary scenario where errors can occur is when the user presses the button to end their turn, but the board is unable to recognize where their piece has moved to. This would result in the user having to reset their piece and make their move again. The second scenario for errors occurs when verifying the piece positions after an automatic move. ChessConnect scans the positions after every automatic move to make sure that piece placement is correct. If pieces become misaligned during movement from either inaccuracy or being knocked too far out of place, the user will have to manually align the pieces until they are properly detected. According to online database chessgames.com, the average chess game is roughly 40 moves, so this specification outlines how many detection errors are acceptable in an average game [5]. The design target for this specification is 1 error per 40 moves, which

allows for some error in piece detection while allowing for minimally interrupted gameplay so that the user experience will be maintained. The threshold of 3 errors per 40 moves was determined in a similar fashion and is viewed as being the upper limit of permissible errors before the user experience starts to diminish.

- **Piece Scaling (King Height):** the measured height in inches of the king chess piece. The relative heights of chess pieces are scaled relative to the height of the king. Many chess players are particular about the scaling of their pieces and may want them to be a certain height in a chess set. The design target is 3.5 inches tall, which is approximately the average height for the king piece in consumer chess sets. The design threshold for this specification is 2.25 inches, which will make sure that the king and piece height is not too small. This ensures that the pieces do not look strange or become difficult to handle.

Table 1: Functional Requirements and Engineering Specifications

Engineering Specifications • Related Functional Requirements	Units	Desired Direction	Design Target	Design Threshold
Manufactured Cost • Affordability • Playable Area • Size of Pieces • Build Quality • Piece Movement Speed	\$	↓	\$350	\$500
Size of Device (L x W x H) • Affordability • Playable Area • Size of Pieces	Inches	↓	27x19x3.5	32x24x4.5
Automatic Piece Move Time • Piece Movement Speed • Affordability • Playable Area • Size of Pieces • Build Quality • Gameplay Options	Seconds	↓	6	11
Time to Start a Game • Ease of Use • Gameplay Options	Screens	↓	3	6

Table 1 cont.

Piece Detection Errors (per 40 moves) • Affordability • Build Quality • Piece Movement Speed • Ease of Use • Gameplay Options	#	↓	1	3
Piece Scaling (King Height) • Affordability • Playable Area • Size of Pieces • Build Quality • Ease of Use • Piece Movement Speed	Inches	↑	3.5	2.25

Competition

The competition for ChessConnect includes several companies that offer similar products aimed at bridging the gap between traditional and online chess. Some of the key competitors include Miko Chess Grand, ChessUp, Chessnut Air +, and DGT Pegasus. These competitors, along with where they rank in terms of functional requirements, are shown in the HoQ in Figure 3. While these products provide the core functionality of electronic chess boards by linking physical and online play, ChessConnect stands out in several key areas.

One of the biggest advantages that ChessConnect has over the competition is the ability to move the opponent's pieces automatically. All the competitors listed except for Miko Chess Grand lack this capability. Additionally, the Miko's price point of \$550 makes it less accessible and affordable. These competitors are also largely training focused, with emphasis on teaching modes, analysis, and puzzles. ChessConnect focuses on reducing complexity by providing users with a fast and straightforward solution for playing online chess games over the board. While competitors may offer the online compatibility of ChessConnect, none of them offer the combination of functionality, convenience, and affordability of ChessConnect.

House of Quality

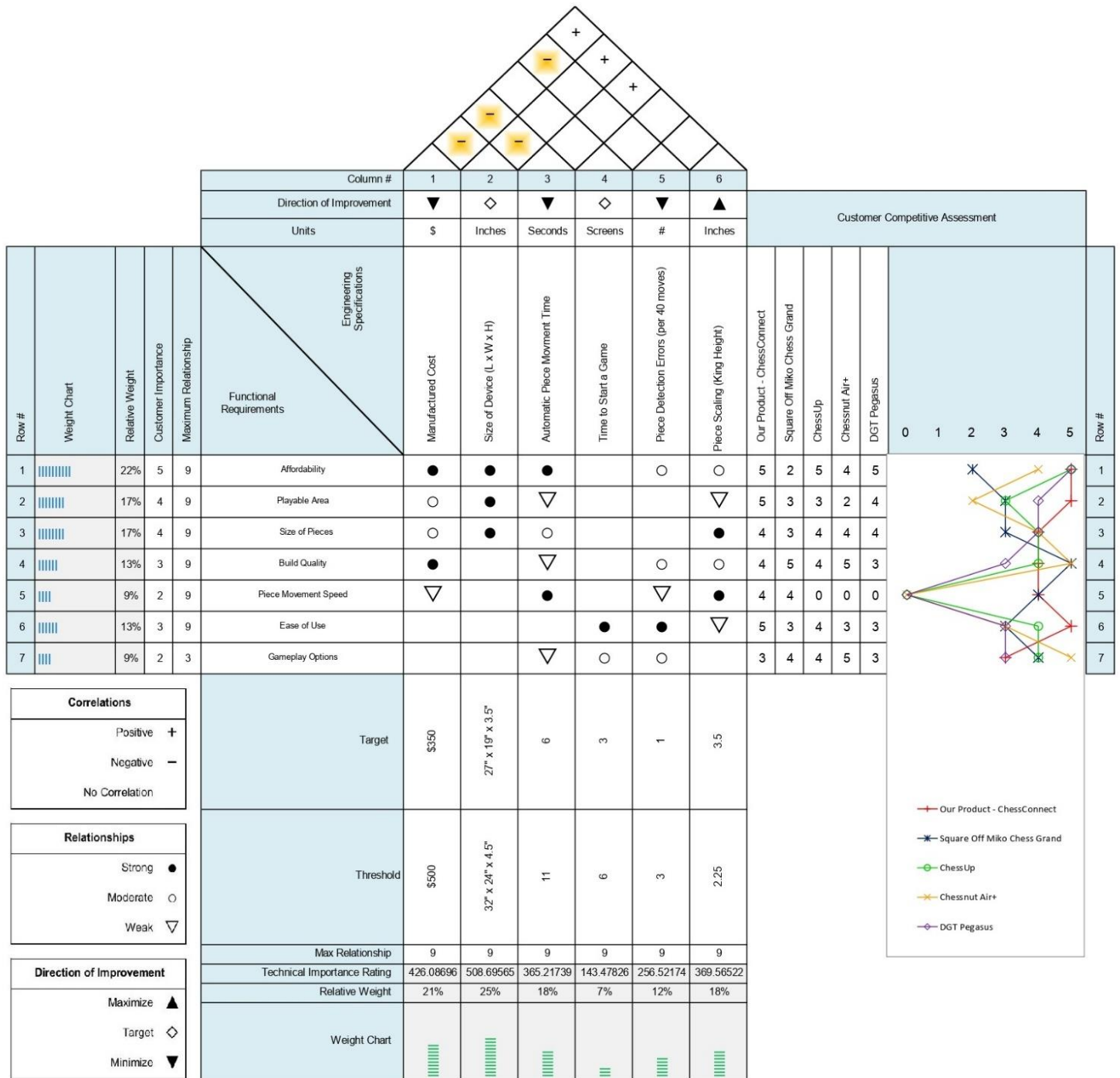


Fig. 3. House of Quality

Impact

ChessConnect aims to bridge the gap between online and over-the-board chess formats. By merging the good habit building of over-the-board chess and the convenience of online chess, ChessConnect creates a unique and enjoyable way for any user to play chess. In a global context, ChessConnect enables players from different parts of the world to compete in a physical environment, while staying connected to the online community. Economically, ChessConnect aims to be affordable and accessible, offering a cost-effective alternative to high-priced competitors without sacrificing quality.

The materials that make up ChessConnect are chosen to reduce its environmental impact as much as possible. 3D printing allows for precision in material usage, reducing more waste than traditional molding or cutting processes. Furthermore, the use of wood to construct the overall device housing as well as other components in future versions (such as the chess pieces) will help to make the product more recyclable. However, the use of plastics in 3D printing raises some concerns, as they can cause harm to the environment if not properly disposed of during the manufacturing process. The energy consumption of ChessConnect can indirectly impact the environment, although this is extremely minimal considering the low power of the device. The electronics inside the device also pose an environmental risk if they are not disposed of properly at the end of the product's life cycle. By mitigating these risks, ChessConnect demonstrates a safe and affordable device that enhances the chess-playing experience while fostering global connectivity and maintaining environmental responsibility.

Safety

ChessConnect is designed to be a consumer-facing product, and therefore ensuring user safety is extremely important. The board requires connection to external power, which introduces risks of electrical shock and overheating. The board should not be operated in the rain or other wet environments to reduce the risk of electronic malfunctions. The continuous operation of the electromagnet, motors, and other electronic components also poses the risk of overheating. Although it is unlikely, the force of a magnetically moved piece being launched off of the board could potentially cause bodily harm. To reduce this risk, the movement of pieces will be calibrated such that these moving parts are not inherently

dangerous. Typical interaction with chess pieces poses no harm, but since the pieces are small, they can act as a choking hazard and could cause pain if stepped on. Therefore, users are advised not to use the chess pieces outside of their intended purpose. Along with risks associated with pieces, the device itself can be hazardous to users. The device can cause bodily harm if a user were to drop the device onto themselves or if they trip and fall over the power cord. Overall, ChessConnect is an extremely safe product when used in the proper context.

Conclusion

ChessConnect offers a unique solution that blends the strengths of both online and over-the-board chess formats; creating a product that enhances the player's experience through accessibility, affordability, and ease of use. ChessConnect focuses on maximizing functionality and consumer needs while still standing out in its market. It not only fosters global connectivity but also builds good chess-playing habits. By addressing safety concerns and acknowledging environmental impact, ChessConnect balances innovation with responsibility. As a result, ChessConnect stands out as a modern, functional, and convenient tool for chess enthusiasts and new players alike.

References

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